The Chinese University of Hong Kong, Shenzhen
Personal study materials · Unofficial

CSC5041 Database Systems

Practice Papers 1 & 2

Questions only · No answer key · 100 marks each

Adapted from the supplied academic report template.

Original teaching materials remain unchanged.

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CSC5041 — Practice Paper 1

Scope: Weeks 1–5 · **Total:** 100 marks · **Suggested time:** 120 minutes

Original practice questions based on the teaching plan and Tutorials 1–2. The time and marks are practice settings, not the lecturer's announced examination format. No answers are included.

Instructions

- Attempt all questions. Label answers by question and subpart.
- Prefer formulas, schemas, diagrams, and SQL. Use at most one or two sentences when an explanation is requested.
- Use the lecture's Chen ER notation. Underline primary keys and mark foreign-key attributes with $*$; state complete composite foreign-key groups and targets.
- Relational algebra uses set semantics. You may use σ, π, ρ, $\cup$, $\cap$, $-$, $\times$, $\bowtie$, $\div$, and γ. Define any intermediate relations.
- Write PostgreSQL-compatible SQL using ordinary SELECT, joins, aggregation, and subqueries. No window functions, recursive queries, or PL/pgSQL are needed.
- Names and titles need not be unique. Unless stated otherwise, all attributes except explicitly nullable ones are non-NULL, and every declared foreign key is valid.
- Each question is independent. Q2's ER model is not the schema used in Q3–Q5.

Q1. Database fundamentals [10 marks]

(a) [2 marks] Distinguish a database from a DBMS. Give one short definition of each.

(b) [3 marks] Name the three-schema level corresponding to each description:

1. A research assistant sees only the projects assigned to that assistant.
2. The complete logical definitions of Scientist, Project, and their constraints.
3. The organization of database records into disk pages and access paths.

(c) [3 marks] Classify each change as demonstrating physical data independence, logical data independence, or neither. Assume any stated interface preservation is achieved successfully.

1. A file organization changes while relation definitions and application queries remain unchanged.
2. A logical relation is split into two, but suitable mappings preserve the existing external interfaces.
3. A scientist's phone number is replaced by a new value.

(d) [2 marks] Classify `CREATE TABLE` and `UPDATE` as DDL or DML, using the course's terminology.

Q2. Conceptual design and relational mapping [30 marks]

A research organization maintains the following information.

A research centre has a unique `centreID`, a name, an address consisting of city and street, and zero or more phone numbers. A centre employs one or more scientists. A scientist has a unique `scientistID`, a name, and a salary, and works at exactly one centre. Each centre has exactly one head, who must be a scientist employed at that centre. A scientist heads at most one centre.

A project has a unique `projectID`, a title, and a budget, and is hosted by exactly one centre. A centre may host zero or more projects. Scientists may work on zero or more projects, and each project must

have one or more scientists working on it. Record `hoursPerWeek` for each scientist–project association. Each recorded value is positive, and a scientist's total `hoursPerWeek` across all projects must not exceed 40. Scientists may work on projects hosted by other centres.

A project may have zero or more experimental runs. Runs are numbered `1,2,3,...` within their project; different projects may use the same `runNo`. Each run has a date and an outcome and is led by exactly one scientist. A scientist may lead zero or more runs. No further condition connects leadership with project membership.

A material has a unique `materialCode`, a name, and a unit cost. A run may consume zero or more materials; a material may be consumed by zero or more runs. Record the quantity consumed for each run–material pair. There is at most one such association per pair.

(a) [5 marks] List entity types and attributes. Identify keys; for each weak entity give its partial key, owner, and full identifier.

(b) [9 marks] Draw the ER diagram, including attributes, relationship attributes, cardinality ratios, participation constraints, and identifying relationships. State any additional assumptions.

(c) [12 marks] Produce the relational schema using the standard mapping. Show all primary keys, foreign keys, and any uniqueness or non-null requirements needed to preserve the stated associations. State which relationships or attributes create separate relations.

(d) [4 marks] Give two stated business constraints not guaranteed by the primary-key and foreign-key declarations in your mapping alone. Express each constraint as a formula or a short sentence, and name a suitable enforcement mechanism.

Q3. Relational integrity [10 marks]

Use the following independent schema:

```
Catalogue(supplierID, partNo, label)
  PK{supplierID, partNo}

Request(requestID, supplierID, partNo, quantity)
  PK{requestID}
  FK{supplierID,partNo} → Catalogue{supplierID,partNo}
  All attributes NOT NULL; quantity is an integer with CHECK(quantity > 0).
```

Initial state:

Catalogue.supplierID	partNo	label
S1	P1	Sensor
S2	P2	Cable

Request.requestID	supplierID	partNo	quantity
R1	S1	P1	2

Assess each operation **independently from this initial state**. No automatic cascades or reference repairs are configured. State **valid/invalid** and the applicable constraint or reason. **[2 marks each]**

1. Insert `(S1,P1,'Replacement sensor')` into Catalogue.
2. Insert `(R2,S1,P2,1)` into Request.
3. Insert `(NULL,S2,P2,1)` into Request.
4. Delete `(S1,P1,'Sensor')` from Catalogue.
5. Change the quantity of Request R1 from 2 to -1.

Schema for Q4 and Q5

```
Trainee(tid, tname, track, level)       PK{tid}
Module(mid, title, area, hours)         PK{mid}
Enrolment(tid, mid, term, score)        PK{tid,mid,term}
Trainer(rid, rname, area)               PK{rid}
Teaching(rid, mid, term)                PK{rid,mid,term}
```

- Enrolment.tid references Trainee.tid; Enrolment.mid and Teaching.mid reference Module.mid; Teaching.rid references Trainer.rid.
- A Teaching tuple states that a trainer teaches that module in that term. A module offering may have several trainers.
- A trainee may take the same module in different terms. `score` is numeric and NULL until assessment. All other attributes are non-NULL.
- Terms are strings such as `'2026F'`. Module hours are positive numbers. Names and titles may repeat.
- "Has taken" means has an Enrolment tuple, whether or not score is available.

Q4. Relational algebra [25 marks]

(a) [3 marks] Return the titles of modules in area `'Data'` with more than 12 hours.

(b) [4 marks] Return the names of trainees who took at least one module taught by trainer `'R7'` in the same term as their enrolment. Consider all terms.

(c) [5 marks] Return the names of trainees who have taken:

1. At least one `'Data'` module or at least one `'Systems'` module, using a set union.
2. At least one module from each of those two areas, using a set intersection.

(d) [4 marks] Return the names of trainees who have never enrolled in any module. Use set difference.

(e) [6 marks] Return the names of trainees who have taken every module in area `'Data'`. Enrolments may be in different terms, and additional modules in other areas are allowed. Assume at least one Data module exists. Use division, or a correct expression using basic operators.

(f) [3 marks] For each area represented in Module, return the area, number of modules, and average number of hours. Use γ.

Q5. SQL [25 marks]

(a) [3 marks] Return `mid`, `title`, and `hours` for Data modules with at least 12 hours. Sort by hours descending, then mid ascending.

(b) [5 marks] Return distinct trainee IDs and names for trainees enrolled in term `'2026F'` in a module taught by trainer `'R7'` in that term. Write the complete joining conditions.

(c) [5 marks] For each trainee who enrolled in at least two distinct modules in `'2026F'`, return `tid`, the

number of distinct modules, and the average available score for that term. Unassessed enrolments count towards the module requirement. Return only groups that satisfy it, sorted by tid.

(d) [5 marks] Return trainee IDs and names of trainees with no assessed enrolment in any term, including trainees with no enrolments. An enrolment is assessed exactly when score is non-NULL. Use NOT EXISTS.

(e) [7 marks] This subpart uses only the following table:

TestScore.attemptID	score
A1	90
A2	80
A3	NULL
A4	80
A5	NULL

1. Give the single result row of the following query. **[3 marks]**

```sql
SELECT COUNT(*), COUNT(score), COUNT(DISTINCT score)
FROM TestScore;
```

2. Give the attempt IDs returned below, and explain the treatment of NULL in one sentence. **[2 marks]**

```sql
SELECT attemptID FROM TestScore WHERE score <> 80;
```

3. Give the attempt IDs returned below, and justify the result in one sentence. **[2 marks]**

```sql
SELECT attemptID FROM TestScore WHERE score NOT IN (90, NULL);
```

End of Paper 1 — 100 marks.

Scope sources: Teaching plan, p. 1 Weeks 1–5 and p. 2 Tutorials 1–2; Tutorial 1, p. 1; Tutorial 2, p. 1. Questions, scenarios, data, timings, and mark allocations in this paper are original practice material.

CSC5041 — Practice Paper 2

Scope: Weeks 1–5 · **Total:** 100 marks · **Suggested time:** 120 minutes

Original practice questions based on the teaching plan and Tutorials 1–2. The time and marks are practice settings, not the lecturer's announced examination format. No answers are included.

Instructions

- Attempt all questions. Label answers by question and subpart.
- Prefer formulas, schemas, diagrams, and SQL. Use at most one or two sentences when an explanation is requested.
- Use the lecture's Chen ER notation. Underline primary keys and mark foreign-key attributes with $*$; state complete composite foreign-key groups and targets.
- Relational algebra uses set semantics. You may use σ, π, ρ, $\cup$, $\cap$, $-$, $\times$, $\bowtie$, $\div$, and γ. Define any intermediate relations.
- Write PostgreSQL-compatible SQL using ordinary SELECT, joins, aggregation, and subqueries. No window functions, recursive queries, or PL/pgSQL are needed.
- Names and titles need not be unique. Unless stated otherwise, all attributes except explicitly nullable ones are non-NULL, and every declared foreign key is valid.
- Each question is independent. Q2's ER model is not the schema used in Q3–Q5.

Q1. Database fundamentals and keys [10 marks]

(a) [3 marks] Account(accountID, email, displayName) has the following rules: accountID and email are each globally unique and mandatory; displayName need not be unique. No other uniqueness rules are given. List all candidate keys, and give one superkey that is not a candidate key.

(b) [2 marks] Classify each operation as changing the schema or only the instance:

1. Insert a new Account tuple.
2. Add a dateOfBirth attribute to Account's definition.

(c) [3 marks] Name the DBMS component responsible for each task:

1. Storing descriptions of table structures and constraints.
2. Choosing a suitable query execution plan.
3. Managing low-level stored-data access and I/O.

(d) [2 marks] A person's address is stored in two independent files. One copy is updated while the other is not. Name the two file-management problems illustrated: the initial duplication, and the disagreement after the update.

Q2. Conceptual design and relational mapping [30 marks]

A lending archive stores the following information.

A publisher has a unique `publisherID`, a name, and an address consisting of city and street. A publisher publishes one or more books, and each book has exactly one publisher. A book has a unique `ISBN` and a title.

Each book has one or more editions. Editions are numbered `1,2,3,...` within their book; the same

`editionNo` may occur for different books. An edition has a publication date and a price. Each edition has one or more physical copies. Copies are numbered `1,2,3,...` within their edition; the same `copyNo` may occur in different editions. A copy records its acquisition date.

An author has a unique `authorID` and a name. An author may write zero or more books, and every book has one or more authors. Record a `contributionPercent` for each author–book association. It is positive, and the percentages for all authors of one book must sum to 100. There is at most one association per author–book pair.

A reader has a unique `readerID`, a name, and zero or more phone numbers. A loan has a unique `loanID`, a checkout date, a due date, and a return date that is NULL while the loan is open. Each loan concerns exactly one reader and exactly one physical copy. Readers and copies may participate in zero or more loans over time. The same reader may borrow the same copy repeatedly, with a different loanID each time. At most one open loan is allowed for a physical copy at any moment.

(a) [5 marks] List entity types and attributes. Identify keys; for each weak entity give its partial key, owner, and full identifier.

(b) [9 marks] Draw the ER diagram, including attributes, relationship attributes, cardinality ratios, participation constraints, and identifying relationships. State any additional assumptions.

(c) [12 marks] Produce the relational schema using the standard mapping. Show all primary keys, complete foreign-key groups, and non-null requirements needed for the associations. State which relationships or attributes create separate relations. Preserve repeated loan events.

(d) [4 marks] Express two stated business constraints not guaranteed by the primary-key and foreign-key declarations in your mapping alone. For each, name a suitable additional enforcement mechanism.

Q3. Relational integrity [10 marks]

Use this independent schema:

```
Desk(buildingID, deskNo, zone)
  PK{buildingID,deskNo}; all attributes NOT NULL

Reservation(resID, buildingID, deskNo, duration)
  PK{resID}
  FK{buildingID,deskNo} → Desk{buildingID,deskNo}
  resID and duration are NOT NULL; CHECK(duration > 0)
```

For this question, the composite foreign key must be either **entirely NULL**, or **entirely non-NULL and matching one Desk key**. This is the lecture convention explicitly adopted for this question, not a claim about every SQL implementation's default.

Initial state:

Desk.buildingID	deskNo	zone
B1	D1	Quiet
B2	D2	Group

Reservation.resID	buildingID	deskNo	duration
R1	B1	D1	2
R2	NULL	NULL	3

Assess each operation **independently from this initial state**. No automatic cascades or reference repairs are configured. State **valid/invalid** and the applicable constraint or reason. **[2 marks each]**

1. Insert `(R3,B1,D2,1)` into Reservation.
2. Insert `(R3,B1,NULL,1)` into Reservation.
3. Change the duration of Reservation R1 to 0.
4. Delete `(B2,D2,'Group')` from Desk.
5. Insert `(B3,NULL,'Quiet')` into Desk.

Schema for Q4 and Q5

```
Customer(uid, uname, city, status)          PK{uid}
Vehicle(vid, model, category, dailyRate)    PK{vid}
Rental(uid, vid, startDate, endDate, charge)
  PK{uid,vid,startDate}
Agent(aid, aname, city)                    PK{aid}
Handles(aid, uid, vid, startDate)           PK{aid,uid,vid,startDate}
```

- Rental.uid references Customer.uid; Rental.vid references Vehicle.vid.
- Handles.aid references Agent.aid. Handles.(uid,vid,startDate) references the complete Rental key.
- A Handles tuple links an agent to one particular rental. Several agents may handle the same rental; an agent may handle several rentals.
- startDate and endDate have DATE type. endDate is NULL for an ongoing rental; charge is numeric and NULL until billing. All other attributes are non-NULL.
- A customer may rent the same vehicle repeatedly on different start dates. Customer and agent cities need not agree.
- "Has rented" means a Rental tuple exists, including ongoing or unbilled rentals. Consider all dates unless a subpart specifies otherwise.

Q4. Relational algebra [25 marks]

(a) [3 marks] Return the models of Hybrid vehicles whose dailyRate is less than 80.

(b) [4 marks] Return the names of customers who have a rental handled by agent `'A2'`. Match the complete rental identity.

(c) [5 marks] Return the names of customers who have rented:

1. At least one Electric vehicle or at least one Hybrid vehicle, using a set union.
2. At least one vehicle from each category, using a set intersection.

(d) [4 marks] Return the names of customers who have never rented any vehicle. Use set difference.

(e) [6 marks] Return the names of customers who have rented every Electric vehicle in Vehicle. Rentals may be on different dates, and vehicles in other categories are allowed. Assume at least one Electric vehicle exists. Use division, or a correct expression using basic operators.

(f) [3 marks] For each category represented in Vehicle, return the category, number of vehicles, and average dailyRate. Use γ.

Q5. SQL [25 marks]

(a) [4 marks] Return vid, model, and dailyRate for vehicles whose dailyRate exceeds the average dailyRate of all vehicles. Use a scalar subquery. Sort by dailyRate descending, then vid ascending.

(b) [5 marks] Return distinct customer IDs and names for customers with a rental beginning on `DATE '2026-10-01'` that is handled by agent `'A2'`. Ensure that the agent is linked to that particular rental, not another rental by the same customer or for the same vehicle.

(c) [5 marks] For each vehicle category rented by at least two distinct customers, return the category, number of distinct customers, and average available charge per rental. Count ongoing and unbilled rentals towards the customer requirement, but use only non-NULL charges in the average. Sort by category. A repeated equal charge is still a separate rental charge, not a value to deduplicate.

(d) [4 marks] Return customer IDs and names for customers who have never rented an Electric vehicle, including customers who have never rented anything. Use NOT EXISTS.

(e) [7 marks] This subpart uses only the following table:

Payment.pid	amount
P1	NULL
P2	20
P3	20
P4	NULL

1. Give the single result row of the following query. **[4 marks]**

```sql
SELECT COUNT(*), COUNT(amount), SUM(amount), AVG(amount)
FROM Payment;
```

2. Give the payment IDs returned below, and justify the result in one sentence. **[2 marks]**

```sql
SELECT pid
FROM Payment
WHERE amount NOT IN (
    SELECT amount FROM Payment WHERE pid IN ('P1','P2')
);
```

3. State the truth value of the following expression. **[1 mark]**

```sql
NOT EXISTS (SELECT 1 FROM Payment WHERE amount < 0)
```

End of Paper 2 — 100 marks.

Scope sources: Teaching plan, p. 1 Weeks 1–5 and p. 2 Tutorials 1–2; Tutorial 1, p. 1; Tutorial 2, p. 1. Questions, scenarios, data, timings, and mark allocations in this paper are original practice material.